



Learning Rate

Learning rate controls the step size during weight updates.

Too High

Model jumps around and fails to learn.

Just Right

Model converges steadily to a good solution.

Too Low

Model learns extremely slowly.

Training Example

How a model gradually improves its prediction over training:

0.40

Before Training

Initial prediction for class: Pass

0.60

After Updates

Prediction improves with weight adjustments.

0.82

Later

Model converges closer to the actual class.

Overfitting in Neural Networks

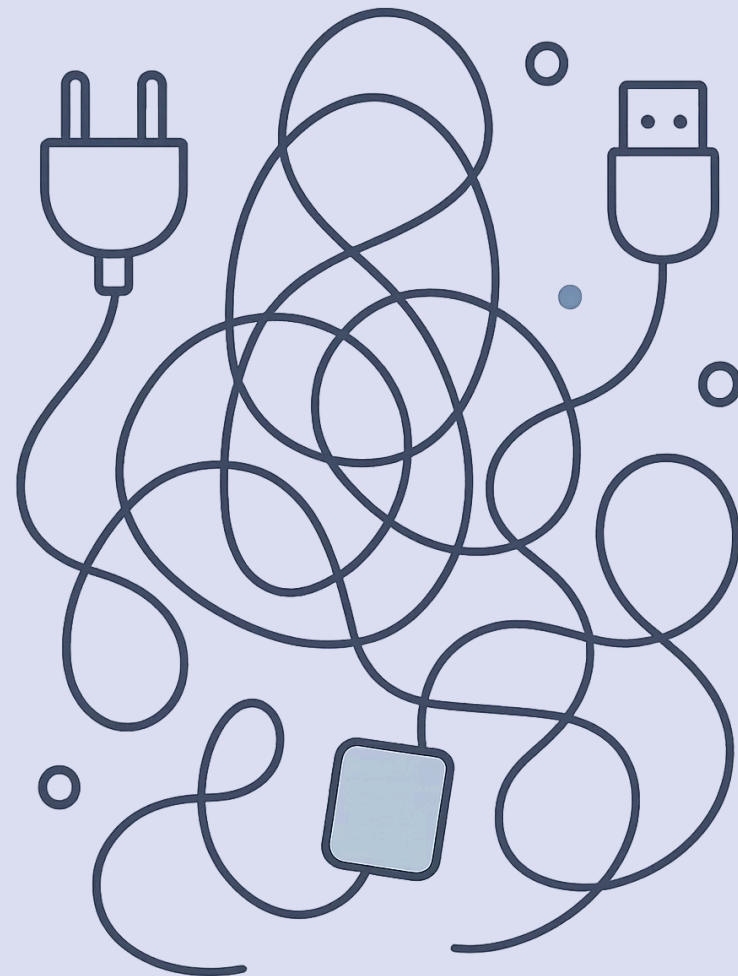
Neural networks can memorize training data rather than generalize.

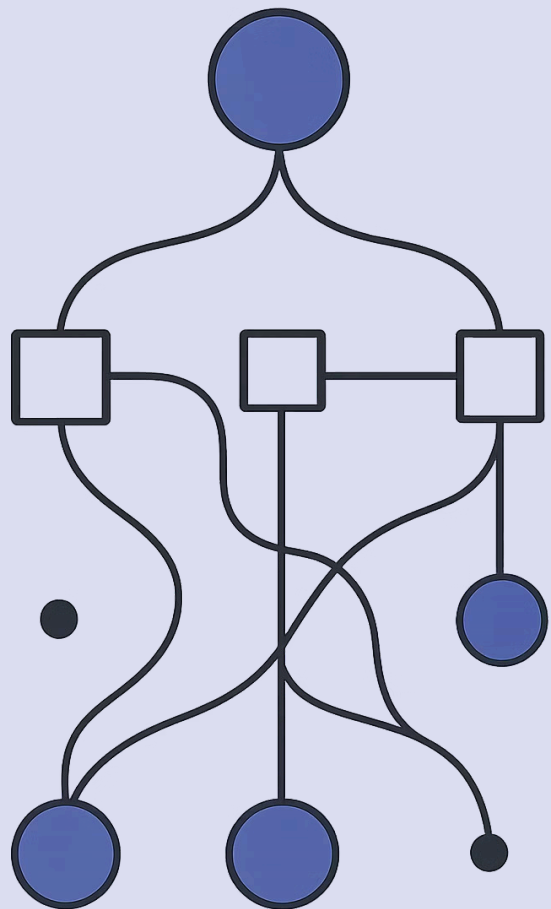
Signs

- Training accuracy: high
- Validation accuracy: low

Fixes

- More data
- Dropout
- Regularization
- Early stopping
- Data augmentation





PART 7

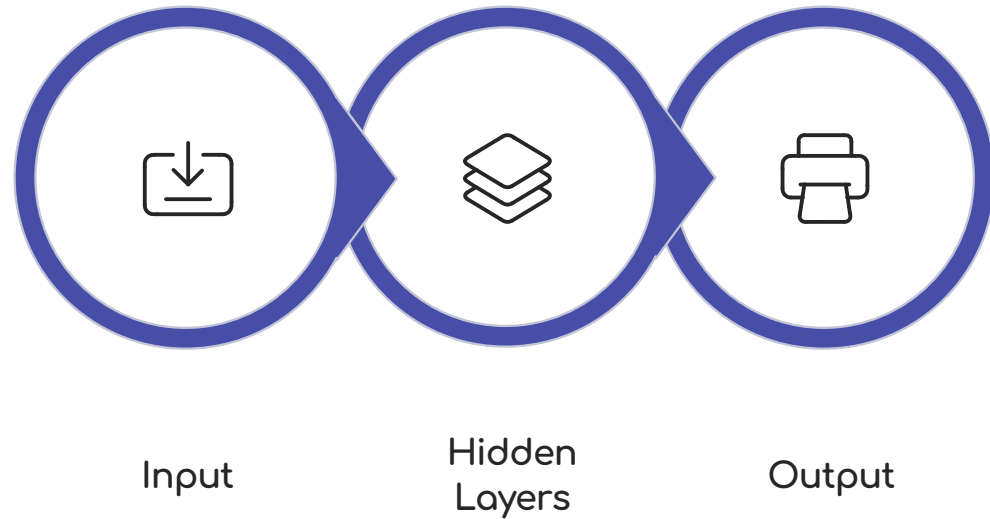
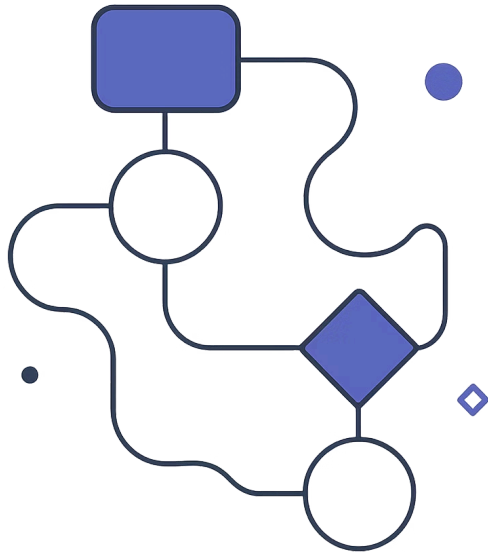
Types of Neural Networks

Neural Network Family Map

Network Type	Best For
ANN / Feedforward	Tabular data
CNN	Images
RNN	Sequences
LSTM / GRU	Long sequences
Transformer	Text and GenAI
Autoencoder	Compression and anomaly detection
GAN	Data generation
GNN	Graph data

ANN / Feedforward Neural Network

ANN is the basic neural network. Data flows in one direction:

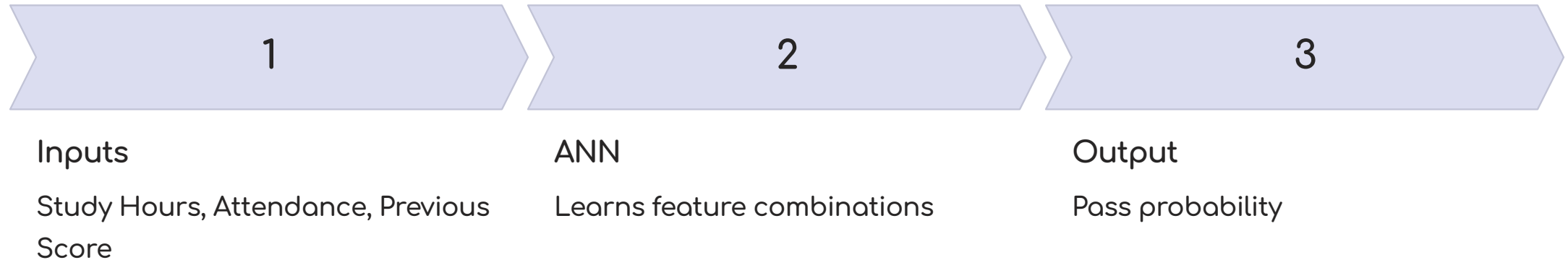


Used For

- Tabular classification
- Tabular regression
- Basic prediction tasks

ANN Example: Student Result Prediction

ANN can learn nonlinear combinations of features to predict outcomes.





CNN — Convolutional Neural Network

CNN is mainly used for image data.

Cat vs Dog

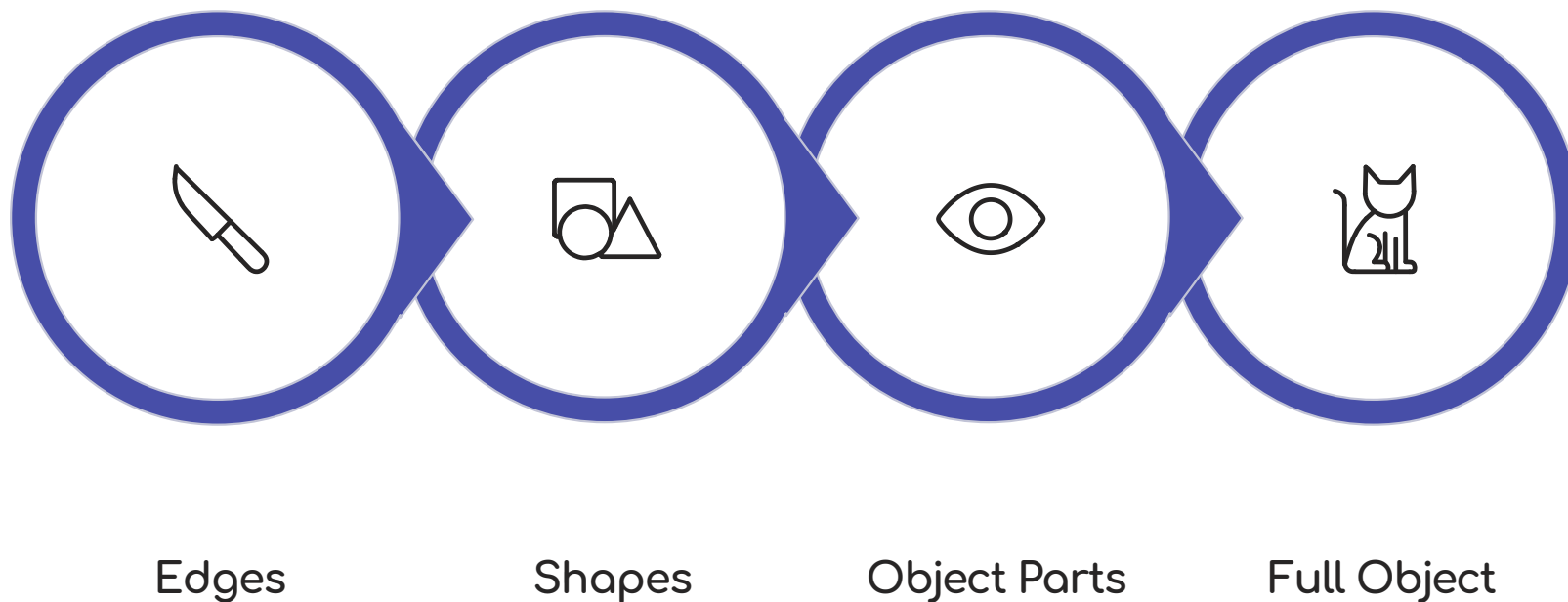
Face Recognition

Medical Scans

Object Detection

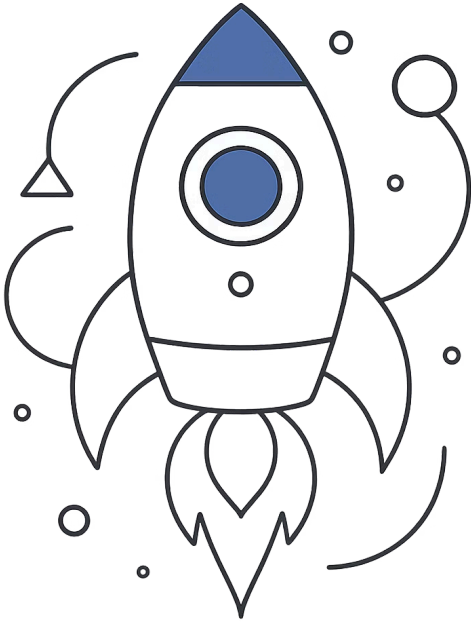
How CNN Works

CNN learns spatial patterns in images, building from simple to complex features.



Each layer detects increasingly complex visual structures, enabling recognition of full objects.

Why CNN for Images?



Spatial Structure

Images have inherent spatial structure — nearby pixels are related.

Local Filters

CNN uses filters to detect local patterns across the image.

- i This spatial awareness makes CNN uniquely powerful for image data.

RNN — Recurrent Neural Network

RNN is designed for sequence data, processing inputs step by step.



Text



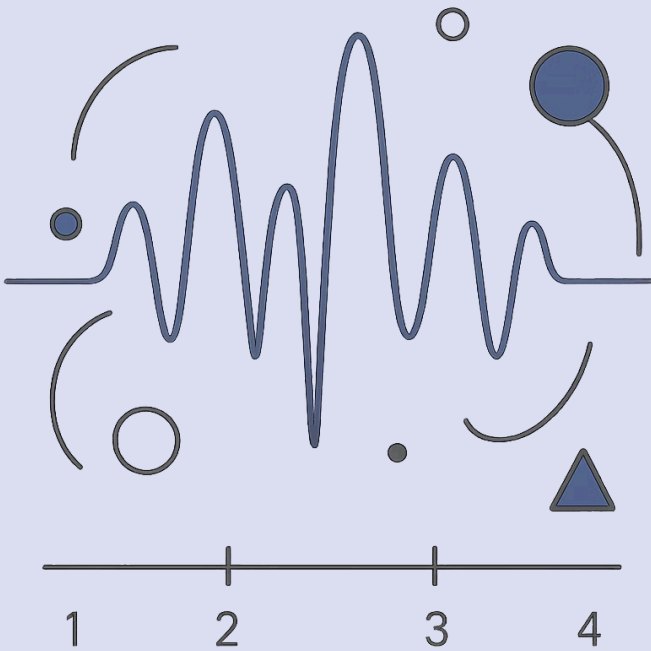
Time Series



Speech



Sensor Data



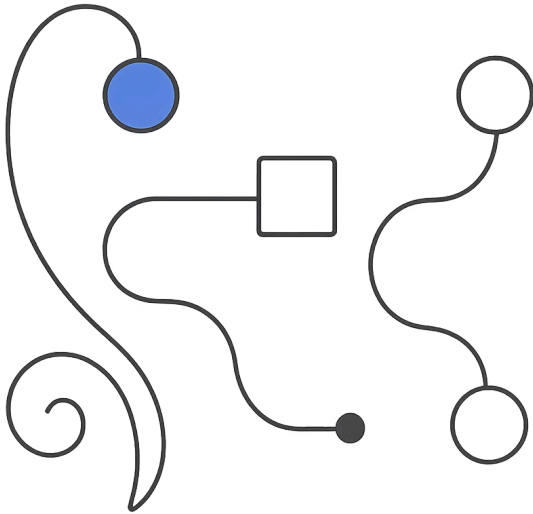
RNN Example

"I loved the movie because it was amazing."

The meaning of later words depends on earlier words. RNN tries to remember previous information as it processes each word in sequence.

- ❑ The word "amazing" only makes sense in context of "loved the movie" — RNN maintains this chain.

LSTM and GRU



LSTM and GRU are improved RNNs that handle longer memory better.

Used For

- Long text
- Speech
- Time series forecasting
- Sequence prediction

Why LSTM / GRU?

Basic RNN forgets long-term context. Consider:

"The movie started slowly, but after the interval it became excellent."

The model must remember both early and later parts of the sentence.

- ✔ LSTM / GRU solve this with gating mechanisms that preserve important context over long sequences.

Transformer

Transformer is the foundation of modern GenAI.

Chatbots

Translation

Summarization

Q&A

Code Generation

Attention Mechanism

Transformers use attention — the model learns which words should focus on which other words.

"The bank approved the loan because it had strong documents."

Attention allows the model to resolve that "it" refers to the bank — not the loan — by weighing word relationships across the full sentence.

Why Transformers Are Powerful

Context Understanding

Understand context well across long passages.

Long Text

Process long text better than RNNs.

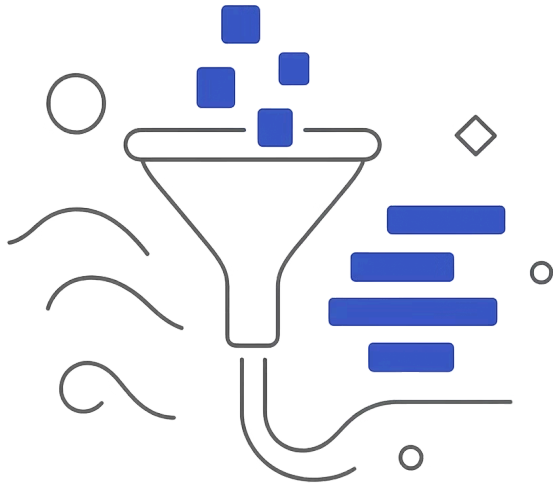
Scalable Training

Train efficiently at scale with parallelization.

LLM Foundation

Power large language models like GPT and Gemini.

Autoencoder



Autoencoder learns to compress and reconstruct data.

Used For

- Dimensionality reduction
- Noise removal
- Anomaly detection

Autoencoder Example: Fraud Detection

An autoencoder trained on normal transactions learns to reconstruct them well.

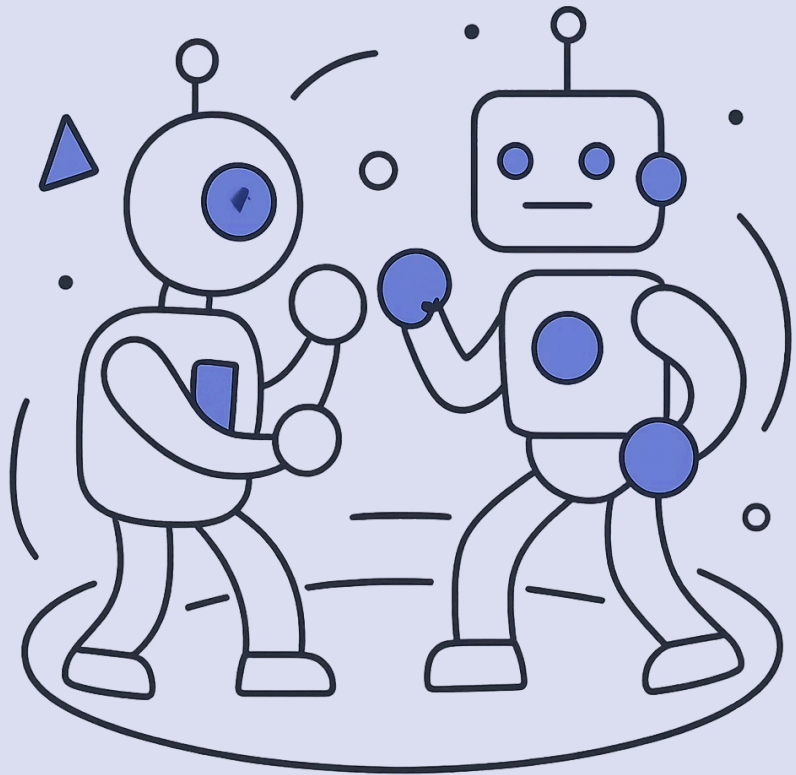
Normal Transaction

Reconstructed well → Low error

Unusual Transaction

Reconstruction error is high

 High reconstruction error may indicate an anomaly or fraud.



GAN — Generative Adversarial Network

GAN is used to generate new data. Two parts compete and improve each other:

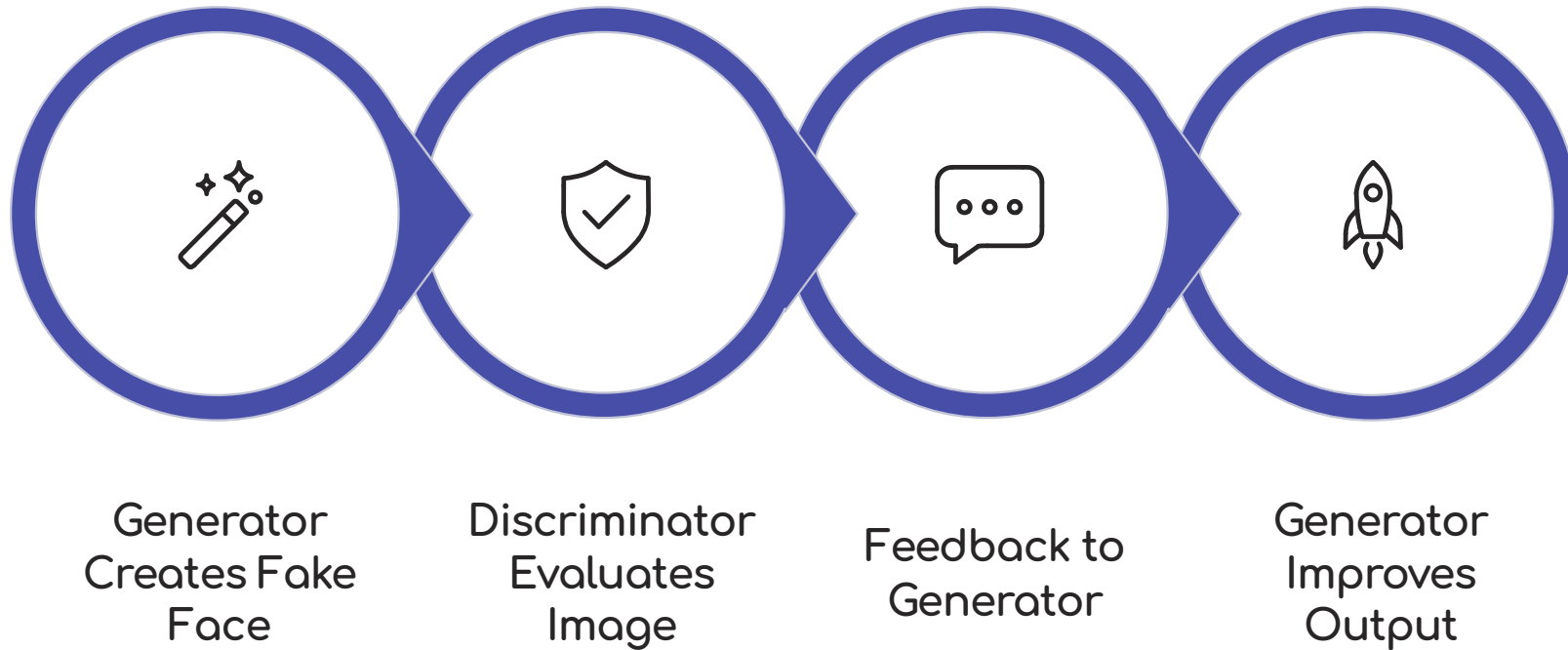
Generator

Creates fake data

Discriminator

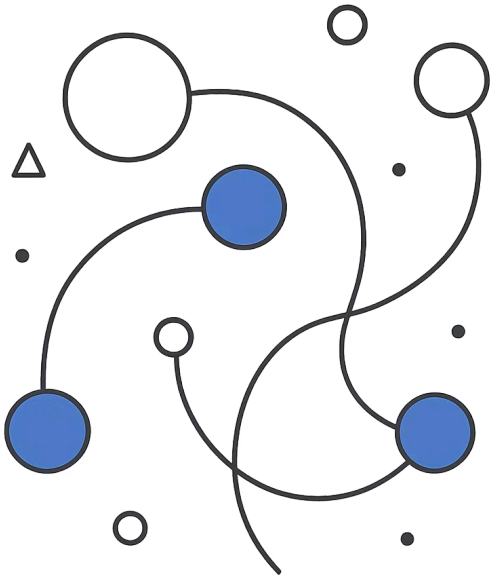
Detects fake vs real

GAN Example: Fake Face Generation



Over many iterations, the Generator becomes increasingly skilled at producing realistic images that fool the Discriminator.

GNN — Graph Neural Network



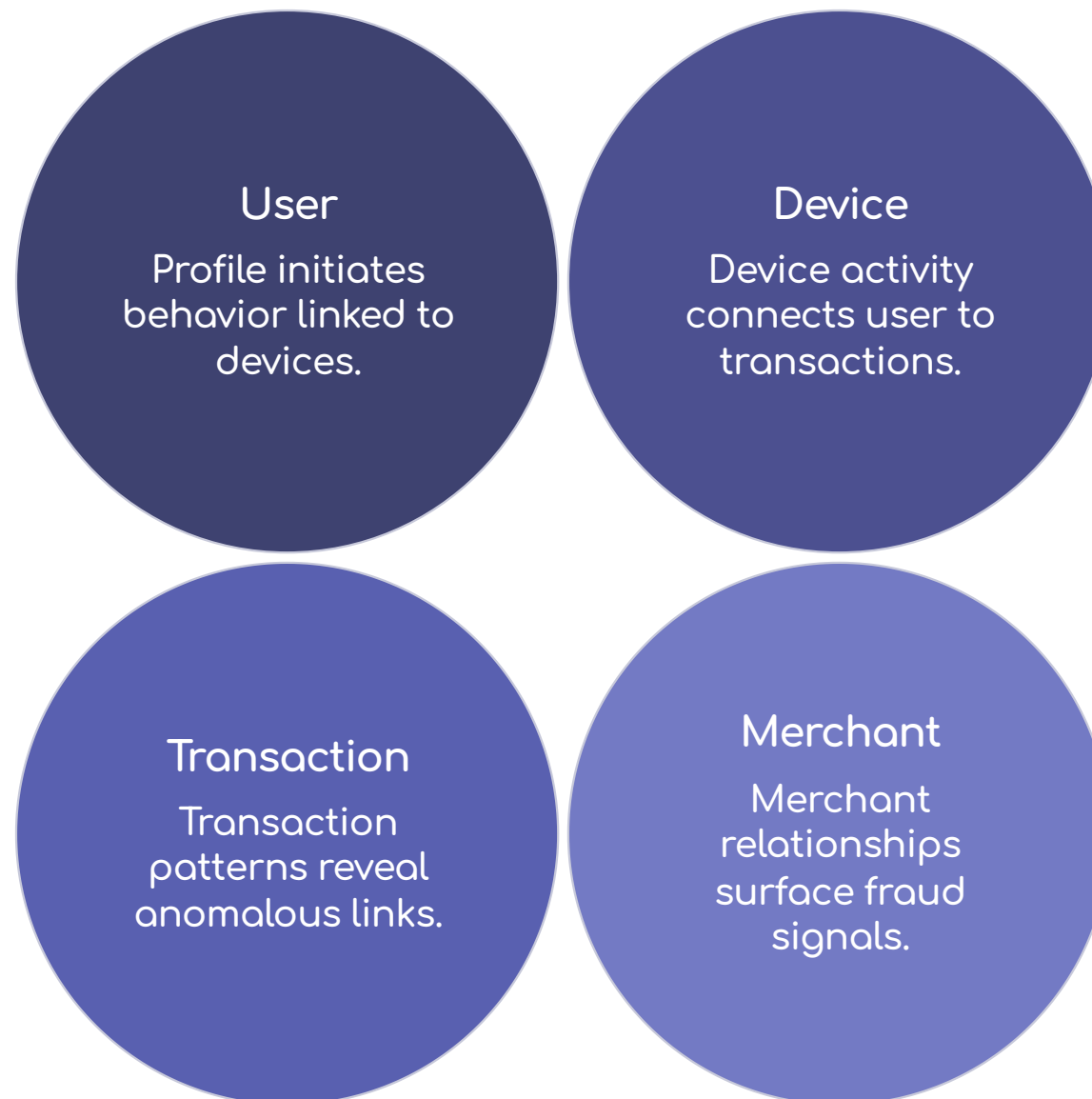
GNN is used for graph data — data with nodes and relationships.

Examples

- Social networks
- Fraud networks
- Recommendation systems
- Molecules
- Knowledge graphs

GNN Example: Fraud Detection

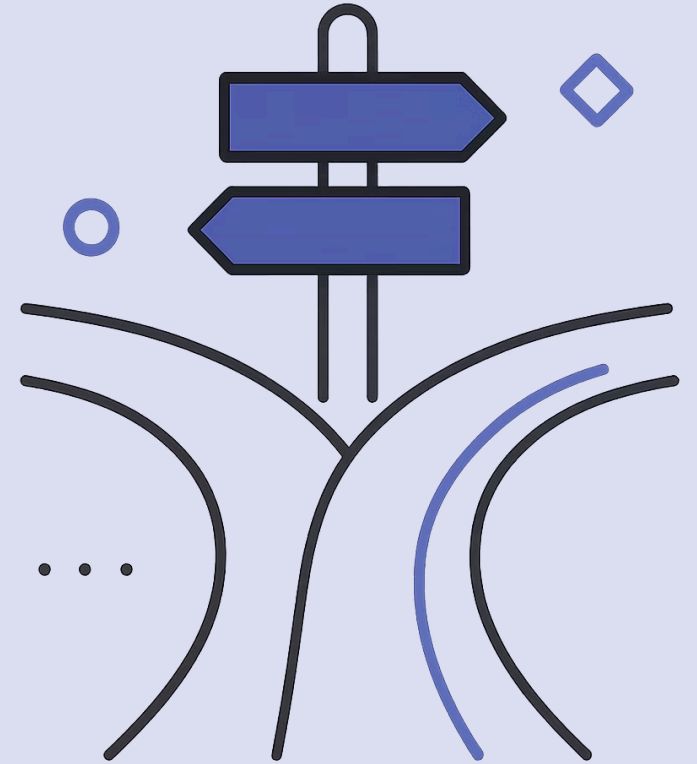
GNN learns from relationships between entities, not just individual records.



⚠ Suspicious connections across this graph may reveal hidden fraud patterns.

PART 8

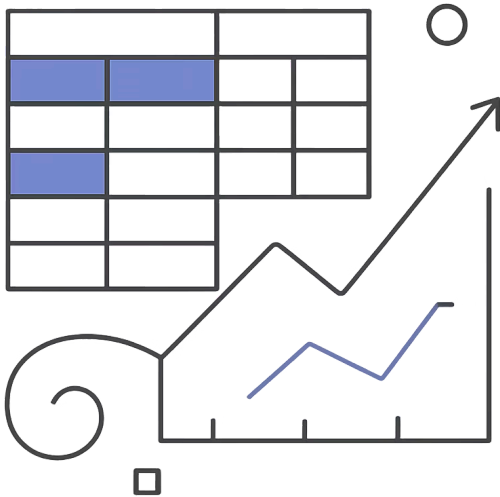
ML vs DL Decision Guide



ML vs DL Comparison

Dimension	Classical ML	Deep Learning
Data type	Tabular	Images, text, audio, video
Feature engineering	Manual	Automatic
Data required	Less	More
Compute required	Lower	Higher
Interpretability	Better	Lower
Training time	Faster	Slower

When to Use Classical ML

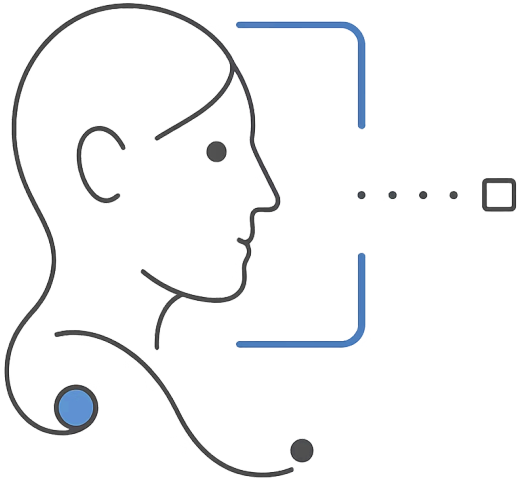


Use Classical ML When:

- Data is structured
- Dataset is small or medium
- Explanation is important
- Training must be fast
- Business needs clear feature impact

 Example: Loan approval with customer table data.

When to Use Deep Learning



Use Deep Learning When:

- Data is raw and complex
- Images, text, audio, or video involved
- Large data is available
- Feature learning is needed
- Highest performance matters

 Example: Face recognition or language translation.

Practical Decision Table

Problem	Better Starting Approach
Student score prediction	Classical ML
Loan default prediction	Classical ML
Cat vs Dog image classification	Deep Learning
Speech recognition	Deep Learning
Customer churn (table data)	Classical ML
Chatbot	Deep Learning

Final Summary

Classical ML

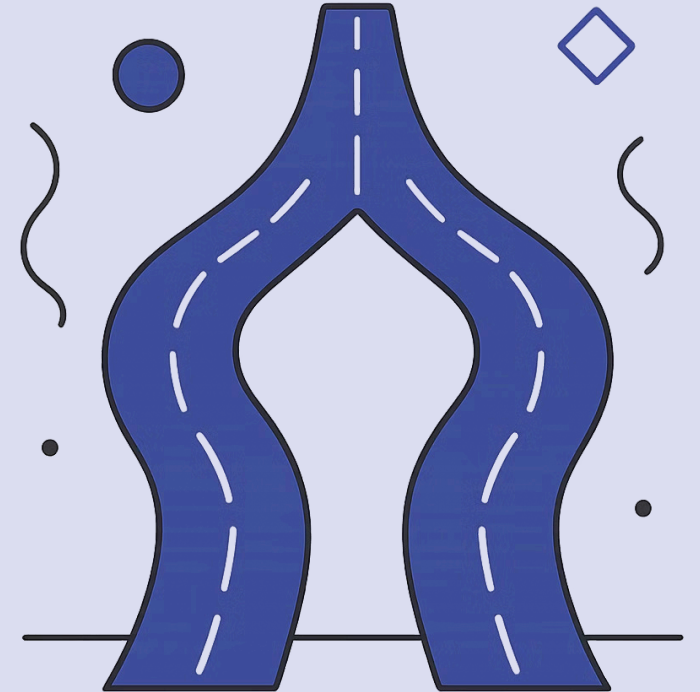
Strong for structured data with clear features.

Deep Learning

Strong for raw, complex data like images and text.

Neural Networks

Learn features automatically — different architectures solve different problems.



Final Takeaway

Classical ML asks:

Can we learn from prepared features?

Deep Learning asks:

Can the model learn useful features by itself?

The right model depends on: data type, data size, accuracy need, compute, and explainability.

NEXT STEP

Deep Learning Hands-On

Build a simple neural network for student pass/fail prediction, then compare:

Logistic Regression

Classical ML baseline

Neural Network

Deep Learning approach

